Department of Computer Science

BS Software Engineering

Course: Calculus and Analytical Geometry

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Mid-Term Examination: Multiple Choice Questions

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1. Which of the following numbers is irrational?
- **(A)** 0.333…
- **(B)** $\sqrt{2}$
- **(C)** 3/7
- **(D)** -5

2. The repeating decimal 0.272727… is equal to
- **(A)** 3/11
- **(B)** 27/100
- **(C)** 2/7
- **(D)** 7/27

3. The fraction 7/40 has a terminating decimal because the prime factors of 40 are
- **(A)** only 2 and 3
- **(B)** only 3 and 5
- **(C)** 2, 3 and 5
- **(D)** only 2 and 5

4. The interval $[-3, 5)$ in set-builder notation is
- **(A)** $\{x : -3 < x \le 5\}$
- **(B)** $\{x : -3 < x < 5\}$
- **(C)** $\{x : -3 \le x \le 5\}$
- **(D)** $\{x : -3 \le x < 5\}$

5. The set $\{x : x \le 2\}$ is written in interval notation as
- **(A)** $(-\infty, 2)$
- **(B)** $[2, \infty)$
- **(C)** $(2, \infty)$
- **(D)** $(-\infty, 2]$

6. If $A = [-4, 2)$ and $B = (1, 6]$, then $A \cap B$ is
- **(A)** $[1, 2]$
- **(B)** $(1, 2]$
- **(C)** $(1, 2)$
- **(D)** $[-4, 6]$

7. If $A = [-4, 2)$ and $B = (1, 6]$, then $A \cup B$ is
- **(A)** $[-4, 6]$
- **(B)** $(-4, 6)$
- **(C)** $[-4, 6)$
- **(D)** $(1, 2)$

8. The value of $|-7| + |3 - 8|$ is
- **(A)** 2
- **(B)** -12
- **(C)** 12
- **(D)** -2

9. The solution of $|x - 2| < 3$ is
- **(A)** $(-1, 5)$
- **(B)** $(-\infty, -1) \cup (5, \infty)$
- **(C)** $(-5, 1)$
- **(D)** $[-1, 5]$

10. The solution of $|x - 2| \ge 3$ is
- **(A)** $(-\infty, -1] \cup [5, \infty)$
- **(B)** $[-1, 5]$
- **(C)** $(-1, 5)$
- **(D)** $(-\infty, -1) \cup (5, \infty)$

11. The distance between the points -3 and 5 on the real line is
- **(A)** 8
- **(B)** 2
- **(C)** -8
- **(D)** 15

12. The solution of $|2x - 3| \le 5$ is
- **(A)** $x \le -1$ or $x \ge 4$
- **(B)** $-4 \le x \le 1$
- **(C)** $-1 \le x \le 4$
- **(D)** $1 \le x \le 4$

13. The solution of $3x - 5 < 10$ is
- **(A)** $x < 5$
- **(B)** $x > 5$
- **(C)** $x < 15$
- **(D)** $x > 3$

14. The solution of $-2x > 6$ is
- **(A)** $x > -3$
- **(B)** $x > 3$
- **(C)** $x < 3$
- **(D)** $x < -3$

15. The inequality $x^2 - x - 6 > 0$ is true when
- **(A)** $-2 < x < 3$
- **(B)** $x < -3$ or $x > 2$
- **(C)** $x < -2$ or $x > 3$
- **(D)** $-3 < x < 2$

16. **The point (−4, 3) lies in the**
    (A) first quadrant
    (B) third quadrant
    (C) fourth quadrant
    (D) second quadrant

17. **Which point lies on the y-axis?**
    (A) (0, −5)
    (B) (−5, 0)
    (C) (2, 2)
    (D) (5, −1)

18. **The distance between P(−2, 1) and Q(4, 9) is**
    (A) 10
    (B) 8
    (C) 14
    (D) √28

19. **The midpoint of the segment joining (−2, 1) and (4, 9) is**
    (A) (1, 5)
    (B) (3, 4)
    (C) (2, 10)
    (D) (−1, 4)

20. **The midpoint of AB is M(3, −2) and A is (7, 4). The point B is**
    (A) (5, 1)
    (B) (−1, −8)
    (C) (1, −8)
    (D) (−1, 8)

21. **The slope of the line through (3, −1) and (7, 7) is**
    (A) 1/2
    (B) 2
    (C) −2
    (D) 4

22. **The slope of a horizontal line is**
    (A) 1
    (B) 0
    (C) undefined
    (D) −1

23. **The slope of a vertical line is**
    (A) 0
    (B) 1
    (C) −1
    (D) undefined

24. **The lines y = 3x + 2 and y = −x/3 + 5 are**
    (A) parallel
    (B) perpendicular
    (C) identical
    (D) neither parallel nor perpendicular

25. **The equation of the line with slope 2 passing through (1, 3) is**
    (A) y = 2x + 3
    (B) y = 2x + 1
    (C) y = 2x − 1
    (D) y = x + 2

26. **The slope of the line 4x − 3y + 12 = 0 is**
    (A) −4/3
    (B) 4/3
    (C) 3/4
    (D) 4

27. **The x-intercept of the line 2x + 3y = 6 is**
    (A) 2
    (B) 6
    (C) 3
    (D) −3

28. **The y-intercept of the line 2x + 3y = 6 is**
    (A) 3
    (B) 2
    (C) 6
    (D) −2

29. **Moving from (1, 2) to (5, 5), the increments are**
    (A) Δx = 3 and Δy = 4
    (B) Δx = 6 and Δy = 7
    (C) Δx = 4 and Δy = 3
    (D) Δx = 4 and Δy = 7

30. **The average rate of change of $f(x) = x^2$ on [1, 3] is**
    (A) 8
    (B) 4
    (C) 2
    (D) 3

31. **A function from A to B assigns to each element of A**
    (A) at least one element of B
    (B) at most two elements of B
    (C) no element of B
    (D) exactly one element of B

32. **The relation {(1,2), (2,3), (3,2), (4,5)} is**
    (A) a one-to-one function
    (B) not a function
    (C) a function with domain {2, 3, 5}
    (D) a function but not one-to-one

33. The relation {(1,2), (1,3), (2,4)} is not a function because

(A) the input 2 has no output
(B) the input 1 has two outputs
(C) the outputs are different
(D) there are only three pairs

34. If $f(x) = x^2 - 3x + 4$, then $f(-2)$ equals

(A) 14
(B) 6
(C) -6
(D) 2

35. If $f(x) = x^2 - 3x + 4$, then $f(a + 1)$ equals

(A) $a^2 + 4$
(B) $a^2 - 3a + 5$
(C) $a^2 - a + 4$
(D) $a^2 - a + 2$

36. If $2^x = 1/8$, then x equals

(A) -3
(B) 3
(C) -8
(D) $1/3$

37. The vertical line test is used to decide whether

(A) a function is one-to-one
(B) a function has an inverse
(C) a graph represents a function
(D) a graph is a straight line

38. The horizontal line test is used to decide whether

(A) a graph is a function
(B) a function is one-to-one
(C) a function is continuous
(D) a graph has an intercept

39. The function $f(x) = x^2$ on $\mathbb{R}$ is

(A) many-to-one
(B) one-to-one
(C) not a function
(D) one-to-many

40. Which relation does NOT define y as a function of x?

(A) $y = x^2 - 4$
(B) $y = x^3 + 1$
(C) $x^2 + y^2 = 9$
(D) $y = |x - 1|$

41. If $f(x) = 2x + 1$ for $x < 1$, x^2 for $1 \le x \le 3$, and $12 - x$ for $x > 3$, then $f(5)$ is

(A) 11
(B) 25
(C) 17
(D) 7

42. For the same piecewise function, $f(3)$ is

(A) 7
(B) 6
(C) 9
(D) 3

43. The graph of $f(x) = |x|$ has the shape of

(A) a parabola
(B) a straight line
(C) the letter V
(D) a semicircle

44. The vertex of the parabola $f(x) = x^2 - 4x + 3$ is

(A) $(-2, -1)$
(B) $(2, 1)$
(C) $(2, -1)$
(D) $(4, 3)$

45. The function $f(x) = x^3$ is

(A) many-to-one
(B) one-to-one
(C) not a function
(D) an even function

46. $(f \circ g)(x)$ means

(A) $g(f(x))$
(B) $f(x) \cdot g(x)$
(C) $f(g(x))$
(D) $f(x) + g(x)$

47. If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(x)$ is

(A) $(2x + 3)^2$
(B) $2x^3 + 3$
(C) $2x^2 + 3$
(D) $x^2 + 2x + 3$

48. If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(g \circ f)(x)$ is

(A) $2x^2 + 3$
(B) $4x^2 + 9$
(C) $2x^2 + 6x + 3$
(D) $(2x + 3)^2$

49. With $f(x) = 2x + 3$ and $g(x) = x^2$, the value of $(f \circ g)(2)$ is

(A) 49
(B) 11
(C) 7
(D) 14

50. With $f(x) = 2x + 3$ and $g(x) = x^2$, the value of $(g \circ f)(2)$ is

(A) 11
(B) 16
(C) 49
(D) 25

51. If $f(x) = x + 5$ and $g(x) = 3x$, then $(f \circ g)(x)$ is

(A) $3x + 15$
(B) $3x + 5$
(C) $x + 15$
(D) $4x + 5$

52. If $f(x) = 1/x$ and $g(x) = x + 2$, the domain of $f \circ g$ is

(A) all real x except 0
(B) all real x except 2
(C) all real numbers
(D) all real x except -2

53. If $f(x) = \sqrt{x}$ and $g(x) = x - 4$, the domain of $f \circ g$ is

(A) $[0, \infty)$
(B) $[4, \infty)$
(C) $(-\infty, 4]$
(D) $(4, \infty)$

54. If $f(x) = 2x - 1$, then $(f \circ f)(x)$ is

(A) $4x - 1$
(B) $2x - 2$
(C) $4x - 3$
(D) $4x^2 - 1$

55. Let $f(a) = a + 1$, $g(b) = b^2 + 2$, $h(c) = 7c - 2$. Then $(g \circ f)(-2)$ is

(A) 3
(B) 7
(C) 1
(D) 6

56. With the same f, g, h, the value of $(g \circ h)(f(3))$, that is, $(g \circ h) \circ f(3)$, is

(A) 26
(B) 50
(C) 678
(D) 28

57. For $f(x) = x^2$ and $g(x) = x + 1$, which statement is correct?

(A) $f \circ g = g \circ f$
(B) $(f \circ g)(x) = x^2 + 1$
(C) $(g \circ f)(x) = (x + 1)^2$
(D) $f \circ g \neq g \circ f$

58. The inverse of $f(x) = 5 - 2x$ is

(A) $(5 - x)/2$
(B) $(x - 5)/2$
(C) $5 + 2x$
(D) $1/(5 - 2x)$

59. The inverse of $f(x) = (x - 4)/3$ is

(A) $3x + 4$
(B) $3x - 4$
(C) $(x + 4)/3$
(D) $3/(x - 4)$

60. The inverse of $f(x) = x^3 + 1$ is

(A) $\sqrt[3]{x - 1}$
(B) $(x - 1)^3$
(C) $\sqrt[3]{(x + 1)}$
(D) $\sqrt[3]{(x - 1)}$

61. The inverse of $f(x) = (x + 2)/(x - 3)$ is

(A) $(x - 3)/(x + 2)$
(B) $(3x + 2)/(x - 1)$
(C) $(3x - 2)/(x + 1)$
(D) $(2x + 3)/(x - 1)$

62. The inverse of $f(x) = \sqrt{(x - 1)}$, $x \geq 1$, is

(A) $x^2 - 1$ for $x \geq 0$
(B) $x^2 + 1$ for $x \geq 1$
(C) $1/\sqrt{(x - 1)}$
(D) $x^2 + 1$ for $x \geq 0$

63. The inverse of $f(x) = x^2 + 3$, $x \geq 0$, is

(A) $\sqrt{(x + 3)}$, $x \geq -3$
(B) $\sqrt{(x - 3)}$, $x \geq 3$
(C) $-\sqrt{(x - 3)}$, $x \geq 3$
(D) $(x - 3)^2$, $x \geq 0$

64. Which function has an inverse on all of $\mathbb{R}$?

(A) $f(x) = x^2$
(B) $f(x) = |x|$
(C) $f(x) = x^3$
(D) $f(x) = 5$

65. If $f(x) = 2x - 3$, then $f^{-1}(5)$ equals

(A) 4
(B) 7
(C) 1
(D) 13/2

66. The domain of f^{-1} is equal to

(A) the domain of f
(B) the range of f
(C) the codomain of f
(D) the zeros of f

67. The graph of f^{-1} is the reflection of the graph of f in the line

(A) $y = 0$
(B) $x = 0$
(C) $y = -x$
(D) $y = x$

68. If $f(x) = 4x - 1$, then $f^{-1}(x)$ is

(A) $1/(4x - 1)$
(B) $4x + 1$
(C) $(x - 1)/4$
(D) $(x + 1)/4$

69. Why does $f(x) = x^2 - 4$ on $\mathbb{R}$ have no inverse?

(A) It is not one-to-one, e.g. $f(-1) = f(1)$
(B) It is not defined at 0
(C) Its range is all real numbers
(D) It is a polynomial

70. The domain of $f(x) = \sqrt{(x - 3)}$ is

(A) $(3, \infty)$
(B) $(-\infty, 3]$
(C) $\mathbb{R}$
(D) $[3, \infty)$

71. The range of $f(x) = \sqrt{(x - 3)}$ is

(A) $[0, \infty)$
(B) $[3, \infty)$
(C) $(-\infty, 0]$
(D) $\mathbb{R}$

72. The domain of $f(x) = 1/(x - 4)$ is

(A) $\mathbb{R}$ except 0
(B) $\mathbb{R}$
(C) $(4, \infty)$
(D) $\mathbb{R}$ except 4

73. The range of $f(x) = (x + 1)/(x - 2)$ is

(A) $\mathbb{R}$ except 2
(B) $\mathbb{R}$ except 0
(C) $\mathbb{R}$ except 1
(D) $\mathbb{R}$

74. The domain of $f(x) = \sqrt{(9 - x^2)}$ is

(A) $[-3, 3]$
(B) $(-3, 3)$
(C) $(-\infty, -3] \cup [3, \infty)$
(D) $[0, 3]$

75. The range of $f(x) = \sqrt{(9 - x^2)}$ is

(A) $[0, 3]$
(B) $[-3, 3]$
(C) $[0, 9]$
(D) $[-3, 0]$

76. The domain of $f(x) = 1/\sqrt{(x - 1)}$ is

(A) $[1, \infty)$
(B) $(1, \infty)$
(C) $\mathbb{R}$ except 1
(D) $(-\infty, 1)$

77. The domain of $f(x) = \sqrt{(x + 2)}/(x - 1)$ is

(A) $(-2, \infty)$
(B) $[-2, \infty)$
(C) $[-2, 1) \cup (1, \infty)$
(D) $\mathbb{R}$ except 1

78. The domain of $f(x) = \sqrt{(x^2 - 4)}$ is

(A) $[-2, 2]$
(B) $(-2, 2)$
(C) $(-\infty, -2] \cup [2, \infty)$
(D) $[2, \infty)$

79. The range of $f(x) = x^2 - 6x + 11$ is

(A) $[3, \infty)$
(B) $[11, \infty)$
(C) $\mathbb{R}$
(D) $[2, \infty)$

80. The domain of $f(x) = 2/(x^2 - 9)$ is

(A) $\mathbb{R}$ except 9
(B) $\mathbb{R}$ except -3 and 3
(C) $\mathbb{R}$ except 3
(D) $\mathbb{R}$

Answer Key

1	**B**	21	**B**	41	**D**	61	**B**
2	**A**	22	**B**	42	**C**	62	**D**
3	**D**	23	**D**	43	**C**	63	**B**
4	**D**	24	**B**	44	**C**	64	**C**
5	**D**	25	**B**	45	**B**	65	**A**
6	**C**	26	**B**	46	**C**	66	**B**
7	**A**	27	**C**	47	**C**	67	**D**
8	**C**	28	**B**	48	**D**	68	**D**
9	**A**	29	**C**	49	**B**	69	**A**
10	**A**	30	**B**	50	**C**	70	**D**
11	**A**	31	**D**	51	**B**	71	**A**
12	**C**	32	**D**	52	**D**	72	**D**
13	**A**	33	**B**	53	**B**	73	**C**
14	**D**	34	**A**	54	**C**	74	**A**
15	**C**	35	**D**	55	**A**	75	**A**
16	**D**	36	**A**	56	**C**	76	**B**
17	**A**	37	**C**	57	**D**	77	**C**
18	**A**	38	**B**	58	**A**	78	**C**
19	**A**	39	**A**	59	**A**	79	**D**
20	**B**	40	**C**	60	**D**	80	**B**